IMPLEMENTATION FOUNDATION

Historical Manuscript Fragment Reconstruction & Evidence Explorer

Project Type	Software Development Project
Milestone	Step 2 — Week 1 Milestone
Application Type	Local, interactive research-support application

1. Project Overview

The Historical Manuscript Fragment Reconstruction & Evidence Explorer is a research-support application for libraries, archivists, museums, and history researchers. It will help users compare incomplete or separated manuscript pages and explore possible page sequences using textual, visual, and descriptive evidence. The system will present candidate arrangements and explain the evidence behind them so that researchers can review and compare the suggestions.

Important limitation: The application will suggest possible page arrangements; it will not claim to recover missing text or prove that a proposed sequence is historically correct. Final interpretation will remain with the researcher.

2. Problem Statement

When manuscript pages become separated, damaged, or incomplete, researchers may need to compare page numbers, text, layout, images, and catalogue descriptions manually. This work can take considerable time, especially when there are many possible page matches or when some clues are missing. The project will provide a structured way to compare available evidence, rank candidate relationships, and display why a sequence may be plausible.

3. Project Goal and Objectives

- Allow a user to add manuscript page images, transcribed text, and available metadata.
- Compare pages using available clues such as page numbers, keywords, approximate text matches, visual layout, image similarity, and catalogue metadata.
- Represent possible relationships between fragments as a graph so users can inspect candidate connections.
- Apply ordering constraints to rule out sequences that conflict with known information.
- Display candidate sequences with evidence and clear indications of uncertainty.
- Evaluate the approach using test cases created from manuscripts whose original page order is known.

4. Intended Users and Organizational Value

The intended users are librarians, archivists, museum staff, manuscript researchers, and historians. The application is intended to reduce repeated manual comparison, make the evidence used in a proposed arrangement easier to inspect, and help researchers compare more than one possible sequence. It is a decision-support tool, not an automated authority on historical truth.

5. Functional Requirements

ID	Area	Requirement
FR-1	Input	Accept manuscript page images and allow related text or metadata to be recorded.
FR-2	Metadata	Store and process details such as manuscript identifiers, page numbers, source collection, and descriptive notes when available.
FR-3	Text comparison	Compare transcribed text and keywords using approximate matching.
FR-4	Visual comparison	Extract and compare image or layout features to identify potentially related pages.
FR-5	Candidate relationships	Create a graph of candidate page-to-page relationships based on available evidence.
FR-6	Ordering	Use known constraints to eliminate impossible or conflicting page sequences.
FR-7	Evidence display	Show candidate arrangements and the clues that support each proposed connection.
FR-8	Evaluation	Compare suggested sequences with known original order in controlled test cases.

6. Proposed Technology Stack

Technology	Planned role
Python	Main programming language for application logic and processing.
Streamlit	Interactive local web interface for uploading, reviewing, and comparing pages.
Pandas and NumPy	Metadata handling and numerical calculations.
OpenCV and scikit-image	Image processing and visual/layout similarity features.
RapidFuzz	Approximate matching of transcribed text and keywords.
NetworkX	Representation and inspection of candidate relationships between pages.
OR-Tools	Constraint-based exploration of possible page ordering.
SQLite	Local storage for page records, metadata, comparisons, and results.

7. Data Sources and Evaluation Plan

The planned data sources are publicly accessible digitized manuscript collections from the British Library and Digital Bodleian. The Internet Archive may be used as an additional source when suitable materials are openly accessible and their use is appropriate. For each selected source, the project will record the available catalogue details and the source of the images.

For evaluation, I will select manuscripts with a known page order and create controlled test cases. For example, I may separate selected pages, hide page numbers, or omit some metadata. The system's suggested order will then be compared with the known original order. Results will be assessed by whether the correct neighboring relationships or overall sequence are recovered and whether the displayed evidence helps a user understand each suggestion.

8. Initial Implementation Approach

- **Step 1 — Prepare data:** Select a small, manageable sample of public manuscript pages and record their known order and metadata.

- **Step 2 — Build the data model:** Define how manuscript records, page images, metadata, comparison scores, and candidate links will be stored in SQLite.
- **Step 3 — Implement comparison features:** Add text matching and basic visual/layout comparison, starting with a small test dataset.
- **Step 4 — Build candidate relationships:** Use comparison results to create a graph of plausible page connections.
- **Step 5 — Explore ordering constraints:** Apply known page-number or sequence constraints and investigate candidate orders.
- **Step 6 — Create the interface:** Use Streamlit to display pages, candidate sequences, and the evidence supporting each suggestion.
- **Step 7 — Test and evaluate:** Compare results against known page order and document limitations.

9. Feasibility, Assumptions, and Risks

The project is planned as a single-student, semester-length software development project. It will run locally on a standard laptop and use Python libraries and a local SQLite database, avoiding a requirement for specialized hardware or proprietary databases. The first implementation will focus on a limited dataset and a small number of measurable comparison methods before adding more advanced features.

- **Incomplete or inconsistent metadata:** The system will use only the clues available for each page and should indicate when evidence is limited.
- **Image quality and layout differences:** Visual comparison may be affected by damage, scanning quality, rotation, or differences in page appearance.
- **Uncertain matches:** Similar text or layout does not guarantee that two pages belong together; results must be presented as candidates rather than facts.
- **Scope control:** The initial prototype will prioritize a small, testable workflow instead of attempting to support every manuscript type.

10. Week 1 Foundation Deliverables

- Define the project problem, goal, intended users, and practical limits.
- List the initial functional requirements and the planned technology stack.
- Identify candidate public data sources and a method for creating test cases with known page order.
- Outline the proposed data-processing workflow and initial evaluation approach.
- Create the required repository milestone folder and store this foundation document as a PDF.

11. Expected Outcome

The expected outcome is a working prototype that accepts a manageable set of manuscript fragments, compares available textual, visual, and metadata clues, proposes possible page sequences, and explains the evidence behind the proposals. The prototype will be evaluated using controlled cases with known original order. Its purpose is to support expert review and make comparisons more organized, not to replace scholarly judgment or invent missing manuscript content.

Document note: This milestone describes the planned foundation and initial implementation approach. Specific features may be refined as the project is tested against the selected public manuscript data and course requirements.